ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO ROBOTICS

PAPER 1 · PRACTICE E

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Laid out like the DHET first paper. Section A (30): multiple choice, true or false, matching and acronyms · Section B (70): PIR sensors, navigation, voltage regulators, CAD and robots in industry, gear motors, wheel speed and battery calculations, a crop-spraying drone scenario and a robot-vacuum program to study.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL calculations and write units with the answers.
6. Write neatly and legibly.

SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 Which component changes the Arduino's small control signal into the large current a motor needs?

A Motor driver B LDR
C Capacitor D LED

1.2 Which ONE of the following is NOT a sensor?

A Ultrasonic sensor B IR sensor
C Servo motor D LDR

1.3 A robot that cleans floors in a home is an example of a/an …

A industrial robot. B service robot.
C military robot. D space robot.

1.4 Which Arduino function sets a pin as an input or an output?

A analogRead() B digitalWrite()
C Serial.print() D pinMode()

1.5 A bumper switch on a mobile robot is a … sensor.

A light B contact (touch) C temperature D sound

1.6 Ultrasonic sensors use sound with a frequency above about …

A 20 kHz. B 20 Hz. C 50 Hz. D 2 Hz.

1.7 Which 3D-printing filament is used for flexible, rubber-like parts such as wheel tyres?

A PLA B ABS C PETG D TPU

1.8 A PWM signal with a 50% duty cycle on a 5 V pin gives an average voltage of …

A 5 V. B 0 V. C 2,5 V. D 1 V.

1.9 The capacity of a robot's battery is usually given in …

A mAh. B mph. C kΩ. D dB.

1.10 Which component protects a transistor from the voltage spike of a relay coil?

A Resistor B Diode
C LED D Push button

(10)

[10]

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Write only 'True' or 'False' next to the question number (2.1–2.5) in the ANSWER BOOK.

2.1 A robot vacuum cleaner is an example of a mobile robot.

2.2 analogWrite() produces a true, smooth analogue voltage on any digital pin.

2.3 A relay module lets an Arduino switch a 230 V device safely.

2.4 A standard hobby servo turns continuously at a set speed.

2.5 Serial.println() can be used to help find errors in a program.

(5)

[5]

QUESTION 3

Choose a description from COLUMN B that matches an item in COLUMN A. Write only the letter (A–J) next to the question number (3.1–3.10) in the ANSWER BOOK.

	COLUMN A		**COLUMN B**
3.1	Bumper switch	A	Runs the program and controls the outputs
3.2	Gear motor	B	Gives a steady 5 V from a higher voltage
3.3	Relay	C	Free-turning wheel that supports the robot
3.4	Lithium-ion battery	D	Detects when the robot touches an object
3.5	Voltage regulator	E	DC motor with a built-in gearbox for more torque
3.6	Wheel encoder	F	Displays messages sent from the Arduino to the computer
3.7	Microcontroller	G	Rechargeable power source
3.8	Chassis	H	Measures how far a wheel has turned
3.9	Caster wheel	I	Switches a high-power circuit with a low-power signal
3.10	Serial monitor	J	Frame that holds all the parts

(10)

[10]

QUESTION 4

Write out the full form of each acronym below.

4.1 PCB

4.2 CAD

4.3 RPM

4.4 PIR

4.5 LCD

(5)

[5]

TOTAL SECTION A: [30]

SECTION B

QUESTION 5

5.1 Explain how a PIR sensor detects movement, and give ONE use of it on a robot. (4)

5.2 Explain the difference between a robot that navigates with sensors and a robot that follows a fixed, pre-programmed path. (4)

5.3 Explain why a voltage regulator is needed in the power circuit of a battery-powered robot. (4)

5.4 Explain what CAD is and how it is used before a robot part is 3D-printed. (4)

5.5 Give FOUR ways in which robots make industry safer for workers. (4)

[20]

QUESTION 6

6.1 Explain how the gear ratio of a gear motor affects the speed and the torque at the wheel. (4)

6.2 A robot wheel with a diameter of 6,5 cm turns at 120 rpm.

6.2.1 Calculate the circumference of the wheel. (2)

6.2.2 Calculate the speed of the robot in cm/s. (2)

6.3 Three LEDs are connected in parallel, each drawing 20 mA. Calculate the total current. (2)

6.4 A 1 500 mAh battery supplies a steady current of 0,5 A. Calculate how long it will last. (2)

6.5 Calculate the power used by a 6 V motor that draws 0,75 A. (2)

6.6 Explain what the mAh rating of a battery tells you. (2)

[16]

QUESTION 7

A farm uses a drone to spray its crops. The drone has four propellers, a GPS module, a downward-facing camera, a spray tank with a pump, and a lithium battery. It flies a route over the field by itself and sprays only where the camera sees that the crops are unhealthy.

7.1 Why does the drone need a GPS module? (2)

7.2 How does the camera help the drone to use less spray? (2)

7.3 Why is battery life a challenge for this drone? (2)

7.4 Give TWO safety or legal points the farmer must consider when flying the drone. (2)

7.5 Give ONE advantage of spraying with the drone instead of by hand. (2)

[10]

QUESTION 8

A team built a simple robot vacuum. Two bumper switches at the front tell the robot when it hits something. Two DC motors are controlled through a motor driver on PWM pins, and a relay switches the suction fan. After 50 bumps the robot stops so that its dust bin can be emptied.

Study the code below and answer the questions that follow.

```
const int leftBump = 2, rightBump = 3;
const int lf = 5, lb = 6, rf = 9, rb = 10;   // motor driver inputs (PWM)
const int fanPin = 7;                        // suction fan relay
int bumps = 0;

void setup() {
  pinMode(leftBump, INPUT_PULLUP);
  pinMode(rightBump, INPUT_PULLUP);
  pinMode(lf, OUTPUT); pinMode(lb, OUTPUT); pinMode(rf, OUTPUT); pinMode(rb, OUTPUT);
  pinMode(fanPin, OUTPUT);
  digitalWrite(fanPin, HIGH);                 // switch on the suction fan
  Serial.begin(9600);
}

void drive(int l, int lBack, int r, int rBack) {
  analogWrite(lf, l); analogWrite(lb, lBack);
  analogWrite(rf, r); analogWrite(rb, rBack);
}

void loop() {
  if (digitalRead(leftBump) == LOW) {          // hit something on the left
    bumps++;
    drive(0, 150, 0, 150);  delay(400);         // reverse
    drive(150, 0, 0, 150);  delay(300);         // turn right
  } else if (digitalRead(rightBump) == LOW) {   // hit something on the right
    bumps++;
    drive(0, 150, 0, 150);  delay(400);         // reverse
    drive(0, 150, 150, 0);  delay(300);         // turn left
  } else {
    drive(180, 0, 180, 0);                      // forward
  }
  if (bumps >= 50) {                            // time to empty the bin
    drive(0, 0, 0, 0);
    digitalWrite(fanPin, LOW);
    Serial.println("Bin full - please empty");
    while (true) {}                             // stop here
  }
}
```

8.1 What is the purpose of the drive() function in lines 15–18, and why does it make the program easier to read? (4)

8.2 Explain what the robot does when the LEFT bumper is pressed (lines 21–24). (4)

8.3 Why are the bumpers set up with INPUT_PULLUP, and what value does digitalRead() give when a bumper is pressed? (4)

8.4 At what PWM duty cycle do the motors run when the robot drives forward? Show your calculation. (4)

8.5 What happens after the 50th bump (lines 32–36)? (4)

8.6 Suggest ONE improvement to the program or the robot, and explain how it would help. (4)

[24]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER